

BATTERY PASSPORT - RESEARCH PROTOTYPE

Not a certified battery passport. Fields are oriented on Regulation (EU) 2023/1542, Annex XIII.

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1. Battery and data-source provenance

Battery ID: ul_pur::UL-PUR_N10-EX9_18650_NCA_23C_0-100_0.5-0.5C_i

Data source: ul_pur

Provenance: ul_pur public dataset, processed by this project

Model: multi-source candidate model (encoder: candidate_v1)

2. State of health

State of health: 84.3 %

90% interval: 79.4 - 89.3 %

Interval method: split conformal, 90% target ($\alpha = 0.1$): one fixed half-width calibrated on held-out batteries of the training pool and applied to every battery; it is an in-domain guarantee only and is not adaptive. Measured capacity checkpoints used: 0.

90% conformal interval calibrated on known sources

3. Expected remaining life

Expected remaining life: not shown

Reason: its nearest known source is ul_pur, which the RUL model was never trained on (it only saw NASA+MIT)

RUL is shown only when the nearest known source is NASA or MIT and the battery is not flagged

4. Cycle count

Cycles available for this battery in this app: 194 Assessed cycle: 194

5. Trust status of this assessment

No distribution shift detected (this is not a guarantee)

Nearest known source: ul_pur score nll_min = -65.74 (flagged if ≥ -5.52)

Stated detection rate: the check flagged 81.9% of batteries from a source it had never seen and wrongly flagged 11.0% of batteries from known sources (leave-one-source-out over 16 sources). It is weakest for mich, NASA, snl. About 1 in 5 unfamiliar batteries are not detected; check against a measured capacity when possible.

Typical SOH error (MAE, points): 1.53 for the nearest known source vs about 5.9 on an unseen source

Guarantee: none - this is a screening check.

Input-sanity check: not run: built-in dataset (the one-class SVM check is applied to uploads)

6. Disclaimer

RESEARCH PROTOTYPE - NOT A CERTIFIED BATTERY PASSPORT. This document is generated by a university research project (CellSense). It is not issued under, and makes no claim of compliance with, Regulation (EU) 2023/1542. The numbers are model estimates with the stated uncertainty and limits; they are not measurements and must not be used for safety-critical, commercial or legal decisions without an independent measured capacity test.

Routing was selected on held-out data, so it is a deployment choice, not an unbiased test result.